

Pizza Sales Analysis using SQL Workbench

A Data-Driven Approach to Understanding Sales
Trends

by: Utkarsh Singh Rawat

Project Overview

📌 This project involves analyzing a pizza sales dataset to derive meaningful insights.

📌 The primary objectives include:

- Identifying popular pizza types and categories.
- Understanding peak order times.
- Calculating total revenue and sales distribution.
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📌 Tools & Technologies Used:

- SQL Workbench for data analysis.
- Excel & CSV Files for data storage.

Datasets Used

The dataset consists of four tables derived from CSV files:

- 1 Orders Table – Contains order details, including order date and time.
- 2 Order Details Table – Stores individual pizzas ordered along with their quantity.
- 3 Pizzas Table – Lists pizza details, such as size and price.
- 4 Pizza Types Table – Provides information on pizza names, categories, and ingredients.

Database Schema & Relationships

The dataset follows a relational structure where tables are connected through common keys.

◆ Key Relationships:

- Each order can contain multiple pizzas, creating a one-to-many relationship between the Orders table and the Order Details table.
- The Pizzas table links to the Pizza Types table, forming a many-to-one relationship.

Table Relationship Diagram (ER Diagram)

Orders Table

order_id	order_date	order_time
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(one-to-many)

Order Details Table

order_detail_id	order_id	pizza_id	quantity
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(many-to-one)

Pizzas Table

pizza_id	pizza_type_id	size	price
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(many-to-one)

Pizza Types Table

pizza_type_id	name	category	ingredients
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A dimly lit background image featuring a pizza on a wooden tray, a wine glass, and a bottle of olive oil. The text is overlaid in a white serif font.

To begin with, I performed some basic analysis on the dataset.

Question 1 : Retrieve the total number of orders placed.

```
• SELECT  
    COUNT(order_id) AS total_order  
FROM  
    orders;
```

Result Grid	
	total_order
▶	21350

Question 2 : Calculate the total revenue generated from pizza sales.

SELECT

ROUND(SUM(order_details.quantity * pizzas.price),
2) AS total_sales

FROM

order_details

JOIN

pizzas ON pizzas.pizza_id = order_details.pizza_id;

Result Grid



total_sales



817860.05

Question 3 : Identify the highest-priced pizza.

```
SELECT
    pizza_types.name, pizzas.price
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
ORDER BY pizzas.price DESC
LIMIT 1;
```

Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	

Question 4 : Identify the most common pizza size ordered.

```
SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
;
```

Result Grid		Filter
	size	order_count
	L	18526
	M	15385
	S	14137
	XL	544
	XXL	28

Question 5 : List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    pizza_types.name, SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.name
ORDER BY quantity DESC
LIMIT 5;
```

Result Grid			Filter Rows:
	name	quantity	
▶	The Classic Deluxe Pizza	2453	
	The Barbecue Chicken Pizza	2432	
	The Hawaiian Pizza	2422	
	The Pepperoni Pizza	2418	
	The Thai Chicken Pizza	2371	

A dimly lit, atmospheric photograph of a dining table. In the foreground, a round pizza with a thick crust, topped with melted cheese, sliced onions, and pepperoni, sits on a wooden pizza board. To the left of the pizza, a wine glass filled with red wine is partially visible. Behind the pizza, a bottle of olive oil with a label that includes the word "Flowers" is standing. In the background, a person's hands are visible, one near the pizza and another further back, suggesting a social gathering. The overall lighting is low and warm, creating a cozy and intimate mood.

Moving to a more detailed level of analysis:

Question 6 : Join the necessary tables to find the total quantity of each pizza category ordered.

```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizza_types.category
ORDER BY quantity DESC;
```

Result Grid			Filter
	category	quantity	
▶	Classic	14888	
	Supreme	11987	
	Veggie	11649	
	Chicken	11050	

Question 7 : Determine the distribution of orders by hour of the day.

```
SELECT
    HOUR(order_time) AS hour, COUNT(order_id) AS order_count
FROM
    orders
GROUP BY HOUR(order_time)
ORDER BY order_count DESC;
```

Result Grid			Filter
	hour	order_count	
▶	12	2520	
	13	2455	
	18	2399	
	17	2336	
	19	2009	
	16	1920	
	20	1642	
	14	1472	
	15	1468	
	11	1231	
	21	1198	
	22	663	
	23	28	
	10	8	
	9	1	

Question 8 : Join relevant tables to find the category-wise distribution of pizzas.

```
SELECT
    category, COUNT(name)
FROM
    pizza_types
GROUP BY category;
```

Result Grid			Filter Rows
	category	count(name)	
▶	Chicken	6	
	Classic	8	
	Supreme	9	
	Veggie	9	

Question 9 : Group the orders by date and calculate the average number of pizzas ordered per day.

```
SELECT
    ROUND(AVG(quantity), 0)
FROM
    (SELECT
        orders.order_date, SUM(order_details.quantity) AS quantity
    FROM
        orders
    JOIN order_details ON orders.order_id = order_details.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Result Grid			 Filter
	round(avg(quantity),0)		
▶	138		

Question 10 : Determine the top 3 most ordered pizza types based on revenue.

```
SELECT
    pizza_types.name,
    SUM(order_details.quantity * pizzas.price) AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizza_types.name
ORDER BY revenue DESC
LIMIT 3;
```

Result Grid			Filter Rows:
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	

A dimly lit background image featuring a pizza on a wooden tray, a glass of red wine, and a bottle of olive oil. The text is overlaid in the center.

In the advanced stage, I focused on deeper revenue trends and business impact analysis:

Question 11 : Calculate the percentage contribution of each pizza type to total revenue.

```
SELECT
  pizza_types.category,
  CONCAT(ROUND(SUM(order_details.quantity * pizzas.price) / NULLIF((SELECT
    ROUND(SUM(order_details.quantity * pizzas.price),
      2)
    FROM
      order_details
    JOIN
      pizzas ON pizzas.pizza_id = order_details.pizza_id),
    0) * 100,
    2),
    '%') AS revenue_percentage
FROM
  pizza_types
  JOIN
  pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
  JOIN
  order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue_percentage DESC;
```

Result Grid		Filter Rows:
	category	revenue_percentage
▶	Classic	26.91%
	Supreme	25.46%
	Chicken	23.96%
	Veggie	23.68%

Question 12 : Analyze the cumulative revenue generated over time.

```
select order_date,  
sum(revenue) over (order by order_date) as cummilative_revenue  
from  
(select orders.order_date,  
sum(order_details.quantity * pizzas.price) as revenue  
from order_details join pizzas on order_details.pizza_id = pizzas.pizza_id  
join orders on orders.order_id = order_details.order_id  
group by orders.order_date) as sales;
```

Result Grid			Filter Rows:
	order_date	cummilative_revenue	
	2015-01-01	2713.8500000000004	
	2015-01-02	5445.75	
	2015-01-03	8108.15	
	2015-01-04	9863.6	
	2015-01-05	11929.55	
	2015-01-06	14358.5	
	2015-01-07	16560.7	
	2015-01-08	19399.05	
	2015-01-09	21526.4	
	2015-01-10	23990.350000000002	
	2015-01-11	25862.65	
	2015-01-12	27781.7	
	2015-01-13	29831.300000000003	
	2015-01-14	32358.700000000004	
	2015-01-15	34343.500000000001	
	2015-01-16	36937.650000000001	

Question 13 : Determine the top 3 most ordered pizza types based on revenue for each pizza category.

```
select name, revenue from
(select category, name , revenue, rank() over(partition by category order by revenue desc) as rn
from
(select pizza_types.category, pizza_types.name,
sum((order_details.quantity) * pizzas.price) as revenue
from pizza_types join pizzas on pizza_types.pizza_type_id = pizzas.pizza_type_id
join order_details on order_details.pizza_id = pizzas.pizza_id
group by pizza_types.category, pizza_types.name) as a) as b
where rn <= 3;
```

Result Grid			Filter Rows:	
	name	revenue		
▶	The Thai Chicken Pizza	43434.25		
	The Barbecue Chicken Pizza	42768		
	The California Chicken Pizza	41409.5		
	The Classic Deluxe Pizza	38180.5		
	The Hawaiian Pizza	32273.25		
	The Pepperoni Pizza	30161.75		
	The Spicy Italian Pizza	34831.25		
	The Italian Supreme Pizza	33476.75		
	The Sicilian Pizza	30940.5		
	The Four Cheese Pizza	32265.700000000065		
	The Mexicana Pizza	26780.75		
	The Five Cheese Pizza	26066.5		

Thank you for your time!

I am happy to answer any questions you may have.

